

Parth Mhaske

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EDUCATION

University of Maryland

College Park, MD

B.S. in Computer Science and Applied Mathematics (GPA: 3.8)

Spring 2028

- **Relevant Coursework:** Algorithms, Data Structures, Advanced Linear Algebra, Probability Theory, Computer Systems, Functional Programming Languages
- **Clubs & Organizations:** Smith Investment Fund, Competitive Programming Club, Club Cycling Team
- **Awards:** IMC Prosperity 4 Round 3 Manual Winner; 2x AIME; Science Olympiad Nationals 3rd; FRC World Championship Autonomous Award; Bitcamp Digital Forensics Track Runner-up

EXPERIENCE

Corsha

May 2026 - Present

Software Engineering Intern

Vienna, VA

- Incoming summer intern focused on backend development of OT Authentication systems.

Architect Labs (Prev. Nuntius YCombinator 'S25)

Nov 2025 - May 2026

Software Engineer

(Remote)

- Designed and implemented custom data-based environments with synthetic data generation and automatic evaluation pipelines.
- Built a custom reward system to quantify LLM model alignment and tool-call correctness in tool-based agentic systems.
- Conducted penetration testing and failure analysis on AI models, uncovering edge cases that informed hardening strategies.

Smith Investment Fund

Oct 2025 - Present

Quantitative Analyst

College Park, MD

- Building to a low-latency C++ crypto HFT platform with live Crypto.com WebSocket ingestion via Boost::Beast.
- Designed polymorphic alpha classes with event-driven signal interfaces for backtestable and live-deployable strategies.
- Defined abstract base class contracts supporting momentum, mean-reversion, and threshold strategies with unified execution.

Department of Astronomy, University of Maryland

Jan 2026 - Present

Undergraduate Researcher

College Park, MD

- Built MSTM-based models of light-particle interactions for aggregated, heterogeneous cosmic dust in circumstellar disks.
- Generated synthetic photometric and polarimetric outputs via Mueller matrix analysis; validated with telescopic observations.
- Developed neural network surrogate models to efficiently search high-dimensional parameter spaces for best-fit dust models.

Computational Social Dynamics Lab, University of Maryland

Jan 2026 - Present

Undergraduate Researcher

College Park, MD

- Developed stochastic simulation models extending Ciampaglia et al. to study popularity-driven bias in online cultural markets.
- Ran large-scale Monte Carlo parameter sweeps across manipulation, heterogeneous-user, and time-varying quality regimes.
- Applied Approximate Bayesian Computation methods to calibrate agent-based models against empirical market ranking data.

MASS Lab, University of Texas at Austin

Apr 2024 - Jan 2025

Research Assistant

Austin, TX

- Researched UAV landing stability under stochastic conditions, invited to present at the 2026 AIAA Aviation Forum.
- Derived coupled high-order ODEs for vertical, pitch, and roll motion using spring-damper dynamics and aerodynamic drag.
- Built parallelized Monte Carlo simulation pipeline with 6.2x speedup, optimizing suspension geometry to 99.67% success.

PROJECTS

Clarify | Python, FastAPI, Neo4j, OpenAI API, Sigma.js

April 2026

- Built multi-format evidence ingestion pipeline parsing emails, PDFs, and chats into a Neo4j knowledge graph.
- Engineered deterministic Cypher contradiction rules (polarity flip, amount mismatch) with artifact-level citations.
- Implemented structured LLM entity extraction and interactive Sigma.js graph visualization; demoed on Enron corpus.
- **Result:** Runner-up, Cipher Tech Digital Forensics Prize at Bitcamp 2026.

NJ High Injury Network Generator | Python, FastAPI, PostgreSQL/PostGIS, React, Docker

March 2026 - Present

- Built full-stack web tool automating high-crash corridor identification for NJ municipalities using Poisson significance testing.
- Designed FastAPI backend with PostgreSQL/PostGIS spatial database and SQLAlchemy ORM; React frontend with Leaflet.
- Containerized via Docker Compose; automated data ingestion from NJ DOT crash records, OpenStreetMap, and Census APIs.

Autonomous Maze Navigating Robot | C++, Arduino

Sep 2023 - Jun 2025

- Implemented cascaded PID (encoder velocity/position; gyro heading) with sensor fusion of odometry and IMU to limit drift.
- Built A* pathfinder with direction-aware turn costs, sequential gate constraints, and wall collision checks to minimize runtime.
- Engineered trapezoidal motion profiles with scaled heading correction, stall detection, and kick recovery at a 1kHz loop rate.
- **Result:** 3rd place at the 2024 Science Olympiad National Tournament.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, HTML/CSS/JavaScript, Go, LaTeX

Libraries: NumPy, SciPy, Scikit-learn, pandas, Matplotlib, TensorFlow

Tools: Git, Jupyter, Linux/Bash, Optuna, REST APIs

Technical Areas: Monte Carlo Simulation, ODE Modeling, PID Control, A* Pathfinding, Logistic Regression, k-means Clustering